

1. WAN vs LAN Comparison

Feature	LAN	WAN
Coverage	Small area such as a home, school, or office.	Large area such as cities, countries, or continents.
Speed	Generally provides high speed.	Depends on the service provider and technology.
Example	Office PCs connected through a switch.	An Ahmedabad office connected to a Bangalore data center.

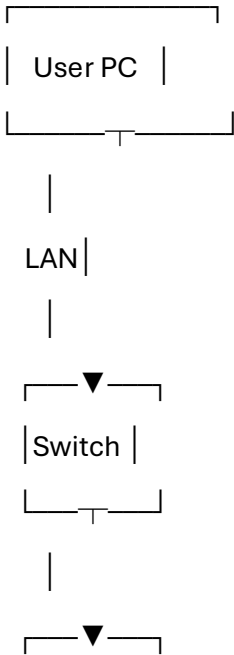
Real-world examples:

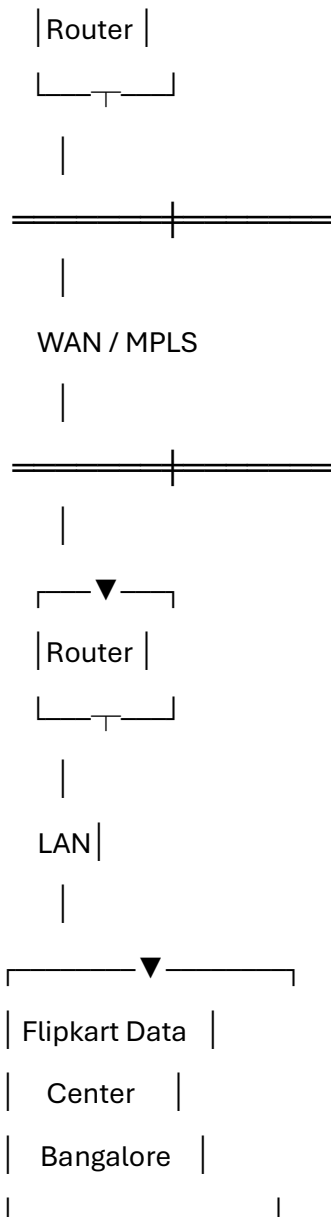
- **LAN:** Devices connected inside an office.
- **WAN:** A user in Ahmedabad accessing a Bangalore data center.

2. WAN Network Diagram

The following diagram shows how a user in Ahmedabad can connect to a Flipkart data center in Bangalore using a WAN.

AHMEDABAD





LAN: PC → Switch → Router

WAN: Ahmedabad Router → ISP/MPLS Network → Bangalore Router

3. What is MPLS?

MPLS stands for **Multiprotocol Label Switching**.

MPLS forwards packets using **labels** instead of relying only on destination IP addresses. It helps improve **traffic management, Quality of Service (QoS), VPN connectivity, and network performance**.

4. MPLS Routing Observations

1. Label-Based Forwarding

MPLS routers use **labels** to forward packets through the network.

2. Label-Switched Path (LSP)

Packets can follow an established path through the MPLS network. The label can change at intermediate routers as the packet travels.

Practical:

The exact labels, interfaces, and routing information should be observed from the provided Packet Tracer/GNS3 MPLS topology.

Screenshot to add: Add a screenshot of the pre-configured MPLS topology or the relevant router output from Packet Tracer/GNS3.

5. Three Benefits of MPLS

1. Traffic Engineering

MPLS helps control traffic paths and reduce network congestion.

2. Quality of Service (QoS)

MPLS can give higher priority to important traffic such as **voice and video**.

3. Scalability and VPN

MPLS helps connect multiple company offices through a scalable WAN and supports **VPN services**.